MIL-A-62463A(AT)
30 March 1984
SUPERSEDING
MIL-S-62463(AT)
2 February 1984

# MILITARY SPECIFICATION

# ACTUATOR ASSEMBLY, ROTARY

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of rotary actuator assemblies unique to the suspension of the Armored Combat Earthmover M9 ACE.

1.2 Classification. Suspension units shall be of the type specified (see 6.2):

| | |
|---|---|
| Type I | - Rotary Actuator Assembly, Corner. |
| Type II | - Rotary Actuator Assembly, Intermediate. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2530

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## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
MILITARY

| | |
|---|---|
| MIL-P-514 | - Plate, Identification, Instruction and Marking, Blank. |
| MIL-L-2104 | - Lubricating Oil, Internal - Combustion Engine, Tactical Service. |
| MIL-L-46167 | - Lubricating Oil, Internal Combustion Engine, Artic. |

STANDARDS
MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-130 | - Identification Marking of U.S. Military Property. |
| MIL-STD-45662 | - Calibration System Requirements. |

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

| | |
|---|---|
| 13204E2560 | - Rotary Actuator Assembly, Corner. |
| 13204E2561 | - Rotary Actuator Assembly Intermediate. |

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.



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## 3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish actuator assemblies which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the assembly to be furnished to the Government. All subsequent assemblies delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes. Any change in production methods after approval shall require a first article resubmittal.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product.

3.2.1 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPL's) whenever available. Contractor's inspection records shall specifically list all QPL items by number and date of QPL, name of supplier and part or drawing number(s). When parts and assemblies are approved as qualified products but not listed on the QPL, the contractor shall list the product by number and date of the approved document and name of supplier(s).

3.2.2 Hydraulic oil. Hydraulic oil used in the actuator assemblies shall conform to grade 10 of MIL-L-2104 for ambient temperatures of -10°F (-23°C) and above and to MIL-L-46167 for 0°F to -65°F (-18°C to -54°C).

3.3 Construction. Actuator assemblies shall conform to the appropriate drawing listed in 2.1.2, and type specified in 1.2.

3.3.1 Interchangeability of parts. Component parts and assemblies of each actuator assembly shall be so constructed that any part(s), may be installed in a like assembly without modification.

3.4 Performance.

3.4.1 Characteristics (Type I). Type I actuator assemblies shall have the following performance characteristics (using MIL-L-2104, 10 WT oil at 90° ± 20°F (32° ± 11°C):

| | | |
|---|---|---|
| a. | Pressures: | |
| | 1. Operating | 6000 pounds per square inch (psi) [41,370 kilopascals (kPa)] |
| | 2. Proof | 15,000 psi (103,425 kPa) minimum |
| | *3. Break-out | 300 psi (2069 kPa) maximum |
| b. | Displacement | 49.6 $in^3$/radian (813 $cm^3$/rad) |
| c. | Torque per 1000 psi (6895 kPa) | 45,000 pound-inch (lb-in) [5084 Newton-meter (N m)] minimum |
| d. | Weight | 260 lb (119.7 kg) dry (Approx) |

* Pressure at which shaft rotation is initiated.



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| | | |
|---|---|---|
| e. | Travel (total) | 72 degrees (°) ± 1° |
| f. | Travel buffing range | 2° to 2.5° at each end of full travel |
| g. | External leakage | None |

1. General [with 6000 psi (41,370 kPa) minimum at vanes]
2. Shaft seal [with 20 ± 5 psi (138 ± 35 kPa) at port 7B)

h. Internal leakage at 3000 psi (20,685 kPa) minimum

| | | |
|---|---|---|
| 1. | Vane seal (both directions of rotation) | 20.3 ounces per minute (oz/min) [300 cubic centimeters per minute ($cm^3$/min)] (maximum) |
| 2. | Hub seal (both directions of rotation) | 20.3 oz/min (300 $cm^3$/min) (maximum) (from Port 7B) |
| 3. | Relief valve [after increasing pressure to 3000 psi (20,685 kPa), then decreasing pressure to 3000 psi (20,685 kPa) following full valve opening] | 2 oz/min (30 $cm^3$/min) (maximum) |
| 4. | Shock check valve | 2.0 oz/min (30 $cm^3$/min) (maximum) |
| 5. | Leveling valve seats | 0.7 oz/min (10 $cm^3$/min) (maximum) |
| 6. | Leveling valve stems | 3.4 oz/min (50 $cm^3$/min) (maximum) |
| 7. | Wheel valve (includes all wheel valve o-rings, and each valve function) | 0.3 oz/min (5 $cm^3$/min) (maximum) |
| 8. | Combined leakage (items 1 + 2 + 6) | 33.8 oz/min (500 $cm^3$/min) (maximum) total (when measured individually) |

i. Leveling valves:

1. The "fill" leveling valve operating range is dependent upon the position of the cam. The valve closes after the wingshaft is rotated in the direction and the amount as shown in table I.
2. The "drain" leveling valve shall open at 9° ± 3° additional rotation from the "fill" leveling valve closing position for all cam positions.
3. Leveling valve plungers to have full freedom of movement in mating bores.

TABLE I. Leveling valve travel

| Cam position | Angle (of travel) | Direction |
|---|---|---|
| 1R | 56° ± 3° | Clockwise |
| 1L | 56° ± 3° | Counterclockwise |
| 4R | 42° ± 3° | Counterclockwise |
| 4L | 42° ± 3° | Clockwise |



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| | | |
|---|---|---|
| j. | Relief valve setting | 4200 ± 100 psi (29,000 ± 690 kPa) cracking pressure |

3.4.2 Characteristics (Type II). Type II actuator assemblies shall have the following performance characteristics (using MIL-L-2104, 10 WT oil at 90° ± 20°F) (32 ± 11°C):

| | | |
|---|---|---|
| a. | Pressures: | |
| | 1. Operating | 6000 psi (41,370 kPa) |
| | 2. Proof | 15,000 psi (103,425 kPa) minimum |
| | 3. Break-out | 300 psi (2069 kPa) maximum |
| b. | Displacement | 49.6 $in^3$/Radian (813 $cm^3$/rad) |
| c. | Torque per 1000 psi (6895 kPa) | 45,000 lb-in (5084 N m) (minimum) |
| d. | Weight | 225 lb (102 kg) dry (Approx) |
| e. | Travel (total) | 72 degrees (°) ± 1° |
| f. | Travel buffing range | 2° to 2.5° at each end of travel |
| g. | External leakage | None |
| | 1. General [with 6000 minimum psi (41,370 kPa) at vanes] | |
| | 2. Shaft seal [with 20 ± 5 psi (138 ± 35 kPa) at Port 7B] | |
| h. | Internal leakage at 3000 psi (20,685 kPa) (minimum) | |
| | 1. Vane seal (both directions of rotation) | 20.3 oz/min (300 $cm^3$/min) (maximum) |
| | 2. Hub seal (both directions of rotation) | 20.3 oz/min (300 $cm^3$/min) (maximum) (from Port 7B) |
| | 3. Combined leakage (items 1 and 2) | 33.8 oz/min (500 $cm^3$/min) (maximum) (when measured individually) |

3.4.3 Temperature. Actuator assemblies shall be capable of operating in ambient temperatures of -65°F to 120°F (-54°C to 49°C) and with oil temperatures of -65°F to 275°F (-54°C to 135°C).

3.5 Identification and marking. Unless otherwise specified on the applicable drawing, each actuator assembly and each major component shall be marked in accordance with MIL-STD-130.

3.6 Nameplate. Unless otherwise specified on the applicable drawing, nameplate, data and instruction plates shall conform to the applicable provisions of MIL-P-514.

3.7 Painting. Actuator assemblies shall be cleaned, treated and painted in accordance with the applicable drawings.

3.8 Workmanship. Workmanship shall be of a quality to assure that actuator assemblies and components are free from defects of a workmanship nature.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection and test requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections or tests set forth in the specification where deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract, the contractor is responsible for the provision and maintenance of all inspection and test equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or contractor designed equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials and construction. To determine conformance to 3.2 through 3.3.1, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
    1. Examination (see 4.5.2).
    2. Acceptance tests (see 4.5.3.1).
    3. Control tests (see 4.5.3.2).

4.3 Inspection conditions and interpretations.

4.3.1 Hydraulic fluid. Hydraulic fluid used for test purposes shall conform to 3.2.2.

4.3.2 Test pressures. Pressures for testing purposes shall be as follows:

a. Operating pressure - 6000 psi (41,370 kPa) (minimum)
b. Proof pressure - 15000 psi (103,425 kPa) (minimum)

4.3.3 Air removal. All entrapped air shall be purged from the actuator assembly and test equipment prior to testing.

4.3.4 Fluid relief ports. The overflow relief ports, specified in 3.4.1 and 3.4.2, shall be vented to an open container for visual examination during all tests. Leakage from the ports defines actual leakage.



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4.3.5 External fluid leakage. External fluid leakage shall be interpreted as any leakage to the outside of the actuator assembly (see 3.4.1.g and 3.4.2.g).

4.4 First article inspection. Unless otherwise specified (see 6.2), the contractor shall furnish one type I and one type II actuator assembly, from the first ten units of each type produced under the production contract, for first article inspection. First article samples shall be examined as specified in table II and tested as specified in table III. Approval of the first article samples by the Government shall not relieve the contractor of his obligation to supply actuator assemblies that are fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval and to accept production units until corrective action by the contractor has been approved by the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all actuator assemblies of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

| Classification | AQL |
|---|---|
| Major | 1.0 |
| Minor | 2.5 |



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TABLE II. Classification of defects.

| Category | Defect | Method of examination |
|---|---|---|
| Major | | |
| 101 | Construction: Not to drawing (see 3.3). | Visual |
| 102 | Dimensions affecting interchangeability: Not within tolerance (see 3.3.1). | Visual and SIE 1/ |
| 103 | Workmanship: Not as specified (see 3.8). | Visual |
| Minor | | |
| 201 | Dimensions not affecting interchangeability: Not to tolerance (see 3.3.1). | Visual and SIE 1/ |
| 202 | Identification and marking: Not as specified (see 3.5). | Visual |
| 203 | Nameplates: Not as specified (see 3.6). | Visual |
| 204 | Painting: Not as specified, poor coverage (see 3.7). | Visual |

1/ SIE = Standard Inspection Equipment

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each actuator assembly shall be subjected to the tests specified in table III.

4.5.3.1.1 Failure. Actuator assemblies that fail to pass any of the acceptance tests specified in table III may be repaired and resubmitted for retest. The retest, as a minimum shall consist of the failed tests and any other tests as deemed necessary by the Government inspector.

4.5.3.2 Control tests. Control tests shall be selected from a production lot which has passed quality conformance examinations specified in 4.5.2 and acceptance tests specified in 4.5.3.1. Control test samples shall be selected at the rate of one from each lot of 300 actuator assemblies consecutively produced, except that not more than one shall be tested in a six month period, nor less than one in a twelve month period. The control test sample shall be subjected to the tests specified in table III.

4.5.3.2.1 Failure. Failure of a sample to pass control testing shall be cause for the Government refusal to accept actuator assemblies until corrective action taken by the contractor has been approved by the Government.



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TABLE III. Classification of tests.

| Title | Requirement | Test | First article | Quality conformance | |
|---|---|---|---|---|---|
| | | | | Acceptance (100%) | Control |
| Characteristics Type I | 3.4.1 | 4.6 | | | |
| | | 4.6 | | | |
| Proof pressure | a.2 | 4.6 | X | | X |
| Break-out pressure | a.3 | 4.6 | X | X | |
| Displacement | b. | 4.6 | X | | X |
| Torque | c. | 4.6 | X | | X |
| Travel | e. | 4.6 | X | | X |
| Travel buffing | f. | 4.6 | X | | |
| External leaks | g. | 4.6 | X | X | |
| Internal leaks | h. | 4.6 | X | | X |
| Level valves | i. | 4.6 | X | | X |
| Relief valve setting | j. | 4.6 | X | X | |
| Charateristics Type II | 3.4.2 | 4.6 | | | |
| Proof pressure | a.2 | 4.6 | X | | X |
| Break-out pressure | a.3 | 4.6 | X | X | |
| Displacement | b. | 4.6 | X | | X |
| Torque | c. | 4.6 | X | | X |
| Travel | e. | 4.6 | X | | X |
| Travel buffing | f. | 4.6 | X | | |
| External leaks | g. | 4.6 | X | X | |
| Internal leaks | h. | 4.6 | X | | X |

4.6 Methods of inspection.

4.6.1 Acceptable methods. The contractor shall develop a series of test methods, acceptable to the Government to assure that actuator assemblies are adequately tested for conformance to all the requirements of 3.4.1 and 3.4.2. The contractor shall use test equipment suited to individual performance requirements (see 4.1.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The actuator assemblies covered by this specification are intended for use as components of the suspension system of the Armored Combat Earthmover M9 ACE.



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6.2 Ordering data. Acquisition documents should specify the following:

a. Type, number, and date of this specification.
b. Type(s) of actuator assembly(s) to be furnished (see 1.2).
c. If first article inspection is not required (see 4.4).
d. Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1).

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| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
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| | ☐ MANUFACTURER |
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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